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EDITORIAL FOCUS

Computational Neuroscience

How sub-optimal are the neural representations: show me your null model

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INTRODUCTION

Neuronal activity in vivo is highly noisy. Both ongoing (1) and stimulus-evoked activity show large within- and across-trial variability (2). Moreover, given the recurrent connectivity and common external inputs, neuronal activity shows small but positive pairwise correlations (3). How neuronal response variability and covariability affect information representation and processing remains one of the important questions in neuroscience.

THE PROBLEM OF NEURAL CODING

Neurons in the brain are tuned to a specific range of stimulus features. For instance, neurons in the visual cortex respond best to a small range of orientations or directions, and neurons in the hippocampus respond best to a specific physical location. Because information is limited by variations in neuronal responses, e.g., spike count variability across trials, it becomes natural to ask how neuronal response variability and covariability may affect the decoding of a stimulus when observing the activity of an ensemble of neurons in a single trial. The answer to this question seems simple enough: when the neuronal activity is uncorrelated, the variance of the stimulus estimate will monotonically decrease as a function of $\sqrt{N}$ (N = number of neurons). On the other hand, when neuronal activity is correlated, averaging across neuronal responses does not reduce the variance of the estimate (4).

When neurons have tuning curves we can calculate the lower bound on variance in the estimate of the stimulus from neuronal activity in terms of the Fisher information (FI):

$$I_F(x) = F'^T(x)\Sigma^{-1}(x)F'(x),$$

where x is the stimulus, $F'(x)$ is the vector of the derivative of the tuning curve, and Σ is the covariance matrix of the neuronal activity. It is important to note that FI is a local measure, in the sense that it only considers small perturbations in the estimate of the stimulus (5) In this setting, covariability (or correlations) in the neuronal activity can be either beneficial or detrimental to decoding the stimulus. To understand this we first need to note that as a population, tuned neurons carve out a stimulus-response curve, representing the average firing rate of the neurons as a function of the presented stimulus (Fig. 1, black line). Geometrically speaking, correlations in the trial-by-trial activity that are orthogonal to the stimulus-response curve (Fig. 1, blue ellipsoid) improve stimulus decoding/representation (6, 7) whereas correlations along the curve degrade decoding (8) (Fig. 1, red ellipsoid).

HOW GOOD IS THE STIMULUS REPRESENTATION IN NEURONAL ACTIVITY?

An interesting question then arises, how optimal or good is the stimulus representation in a brain region or even in the retina? To answer this we need a criterion for optimality, e.g., we can ask: is the stimulus-related information in the neuronal activity maximal given constraints on the sparsity of the firing activity? Importantly, trial-to-trial variability always degrades information representation (mutual information is maximized without noise). To determine how good the representation/decoding is, we need to compare the experimentally measured Fisher information (FI_{data}) with that calculated from a reference or a "null model" (FI_{null}). Thus, any statements regarding the role of correlations on information representation are with respect to the null model.

The choice of a null model is largely a subjective decision. For instance, we can consider an ensemble with independent trial-to-trial variability where the firing rate of each neuron is constrained by the data as a null model. In this model, any correlations among neuronal spiking activity are ignored. If $\mathrm{FI}_{data} = \mathrm{FI}_{null}$, we can argue that the experimentally recorded neuronal activity is either uncorrelated or the correlation structure is such that they do not affect decoding. By contrast, $\mathrm{FI}_{data} < \mathrm{FI}_{null}$ would suggest suboptimal decoding/representation and suggest that correlations impair the stimulus representation. Similarly, $\mathrm{FI}_{data} > \mathrm{FI}_{null}$ implies that the correlations are beneficial. However, different choices of null

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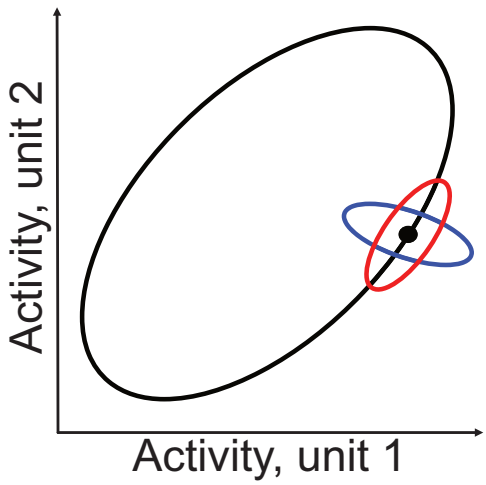

Figure 1. Stimulus-response curve and shape of trial-by-trial activity. By varying the presented stimulus, the mean responses of neurons, i.e., their tuning curves, create a trajectory in activity space (black line). For a given stimulus (black, filled circle), the orientation of the collective trial-by-trial covariability causes differential effects on the uncertainty of the underlying stimulus. Variability along the stimulus-response curve (red ellipsoid) causes a larger uncertainty in estimating the stimulus compared to co-variability oriented orthogonal to the curve (blue ellipsoid).

models may lead to different results. This underscores the important role null models play in measurements, especially if claims are made regarding optimality.

SUMMARY OF THE MAIN RESULTS OF THE PAPER

In this journal, Livezey et al. (9) revisit the question of the optimality of stimulus representation in the retina, V1, and primary auditory cortex. For measuring the optimality of neural coding Livezey et al. (9) have devised two new null models. The typical choice for a null model is Poisson-type spiking, which assumes that there are no correlations among neuronal activity (Shuffle model). However, neuronal activity in early sensory areas shows weak positive correlations. Such correlations naturally arise due to shared inputs and recurrent connections and cannot be completely canceled. Because independent trial-to-trial variability is not always optimal, the authors argue that a null model also should incorporate correlated activity. To account for correlations, the authors introduce the uniform correlation (UC) model which assumes that the pairwise correlations are uniformly distributed. If we visualize neuronal activity as an N-dimensional ellipsoid, the UC model samples ellipsoids of different eccentricities and angles with respect to the encoding trajectory (Fig. 2 in Ref. 9). The second model authors introduce is the factor analysis (FA) model, where private and shared variability of neuronal activity are separated, and rotation of the shared components is sampled from a uniform distribution (Fig. 2 in Ref. 9). Thus, UC and FA models explicitly take into account that neuronal activity has non-zero correlations. The formulation of these models is quite elegant and an important contribution to the paper.

Using these models, the authors found that the median FI for UC and FA is much higher than that of the Shuffle model. In other words, there exist correlation structures that benefit decoding. Second, they found that in their data all FI_{data} [for the neuronal activity in mice retina, primary visual cortex (V1) of macaque and primary auditory cortex (PAC) of rats] are very similar to $\mathrm{FI}_{Shuffle}$, suggesting correlations in the neuronal activity do not contribute much to the decoding. But importantly, FI_{data} is much smaller than both FI_{UC} and FI_{FA} for all three data sets and population sizes (Fig. 3 in Ref. 9). Furthermore, the difference in median FI grows with population size, making the observed correlations more suboptimal.

When the two models are constrained by single-unit neuronal activity, $\mathrm{FI}_{UC} > \mathrm{FI}_{FA}$. Moreover, the distribution of FI_{UC} is heavy-tailed presumably because the UC model can sample any arbitrary level of correlations. In fact, the authors show that high values of FI_{UC} involve very high correlations (near one). In the FA model, pairwise correlations do not reach such high values; however, they assume that correlation ellipsoids can be rotated arbitrarily with respect to the stimulus-response curve.

HOW TO DESIGN A NULL MODEL?

One of the key points the authors make is that null models should be biologically interpretable. The Shuffle model assumes that neurons are neither mutually (effectively) connected nor driven by correlated inputs. This model makes sense only when we want to compare the brain to an idealized ensemble spiking in a Poisson manner. However, we certainly cannot deny that the neurons in the brain are connected following certain rules. So it may be a better question to ask how good the neural representations in the brain could be given its biological constraints. The two null models could be interpreted in terms of the underlying connectivity and inputs. In the FA model, the shared component of the covariability is randomly rotated. In terms of network structure, this would mean altering the underlying recurrent synaptic connectivity. But there are other mechanisms that may control the shared covariability e.g., attention. The UC model is difficult to interpret, however, in some cases, it may imply constructing a neuronal network for which synaptic weights are drawn from a uniform distribution.

In addition to the UC and FA models, other null models can also be formulated, for instance, by randomly shuffling the correlation matrix, Σ. This will ensure that correlations remain identical to what has been measured but by shuffling we will destroy the specific geometry of the correlations.

ARE NEURAL REPRESENTATIONS IN AWAKE BEHAVING ANIMALS ALSO SUBOPTIMAL?

Given the constraints of neuronal connectivity, and excitation-inhibition balance, neuronal populations are unlikely to reach very high correlations (10), and arbitrary rotation of the neuronal activity (both encoding trajectory and the noise correlation) is nontrivial. It seems reasonable to conclude that with respect to the UC and FA models, neuronal representations in the early sensory networks of the brain are highly suboptimal; but it is an open problem whether and how the stimulus-response curve and noise correlation can be independently altered/rotated as both depend on the external input and recurrent connectivity (11, 12). However, we should note that the data used in this study comes from either in

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vitro preparation (retina) or anesthetized animals. Moreover, the use of electrocorticogram (ECoG) signals for this analysis is interesting but questionable since we do not know how many neurons are contributing to any given electrode. Therefore, it is important to verify these results using data from awake behaving animals and not just those passively observing the inputs. In any case, the authors in this paper have provided a powerful framework to address the question of optimality of neural representation in the wake of correlated activity.

DOES IT MATTER WHETHER THE BRAIN IS OPTIMAL OR SUBOPTIMAL?

From many different perspectives (e.g., Refs. 9, 13, and 14) it seems that stimulus representation in the brain is suboptimal. This is inconsistent with the idea of efficient coding. However, this perspective has emerged by assuming that stimulus decoding is done with spike rates (or counts). Alternative coding schemes may not suffer from the same suboptimality. Even within the spike rate coding regime, these studies often use data when animals are either anesthetized or just passively viewing the inputs. Besides, natural stimuli have an underlying temporal and spatial structure, which might be used to improve decoding, for example by using posterior distributions in Bayesian inference or top-down inputs.

Irrespective, the discussion about the optimality of neural representations is very interesting. However, we should remember that there is no single perfect null model. Livezey et al. (9) highlight how our intuition about brain function is influenced by the null models we use, whether explicitly or implicitly. As each null model also acts as a measuring tape, this study shows the importance of developing new null models that are both relevant for brain function and biologically interpretable.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

M.L. and A.K. prepared figures; drafted manuscript; edited and revised manuscript; and approved final version of manuscript.

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